Partner-Specific Affective Precision in Social Active Inference

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Abstract. In multi-agent social settings, model reliability varies across relationships. Beyond inferring what others will do, an agent must calibrate how confidently those inferences should guide action for each relationship. An agent may carry a validated model of one partner, a fragile model of another, and a revising model of a third; collapsing these into a single confidence estimate loses action-relevant information. Affect is often described in terms of valence and arousal, but it can also serve a functional role: regulating how strongly beliefs are expressed in action rather than shaping their content. Using active inference, we model affect as a metacognitive estimate of confidence in the current partner model, formalized as relationship-specific affective precision. Each partner's behavioral evidence updates a confidence estimate that modulates policy precision during selection.

Simulations in a multi-partner graded trust game show that partner-local affective precision acts through policy sharpening rather than improved partner-state inference. Because the precision update is driven by partner-response predictability rather than realized payoff, the mechanism supports model calibration rather than reward tracking. In graded decision settings, the same beliefs produce more committed policies when partner-local confidence is high, without guaranteeing higher payoff. Under abrupt betrayal, confidence built from a reliable relationship can continue sharpening commitment after the partner changes, so adaptation depends on redirecting engagement rather than confidence alone. Varying precision gain and prior model fitness reveals trust-calibration profiles that generate candidate computational hypotheses about individual differences in social trust calibration. Together, these results describe affective precision as a relationship-specific metacognitive signal separating social prediction from social commitment.

Keywords: Active inference · Affective precision · Social metacognition · Multi-agent systems · Trust · Individual differences

1 Introduction

Social decision-making often unfolds across relationships that differ in reliability. In repeated cooperative exchange, the same agent may interact with a familiar collaborator, an uncertain newcomer, and a partner whose behavior is predictable but adversarial. What matters is therefore not only whether a partner is valuable, but whether the agent has a reliable model of how that partner tends to behave. Repeated social exchange depends on learning such partner-specific regularities, including reciprocity, reputation, moral character, and social value [17, 3, 7]. In these settings, the question is not only what an agent believes about each partner, but how confidently those beliefs should guide policy inference.

This separates social decision-making into three related computational problems. The first is partner inference: learning what a partner is like, including whether their behavior is cooperative, exploitative, reciprocal, or unstable. The second is theory of mind: modeling what another agent knows, intends, or expects, and how those representations guide their actions [11, 12, 21]. The third is confidence calibration: estimating whether the focal agent's current partner model is reliable enough to guide policy commitment. Accurate partner inference and sophisticated mentalizing do not by themselves solve this third problem. Knowing that someone is predictable is not the same as knowing whether they are desirable; knowing that someone is unreliable is not the same as knowing how cautiously to act toward them. An agent that infers accurately but commits poorly wastes its social knowledge, while one that commits confidently on fragile models exposes itself to exploitation.

We model social affect as a metacognitive estimate of whether the focal agent's current partner model warrants confident action. This treats affect not as a direct readout of another agent's intentions, nor as a record of reward and punishment, but as a regulator of how strongly current beliefs are expressed in policy. In psychology and neuroscience, affect is often described in terms of valence and arousal—the felt pull toward or away from situations and social commitments [2]. Functionally, however, affect can also assign action-relevant weight to what an agent already infers, shaping the degree of commitment rather than the content of inference.

The active inference framework [9, 10, 19] provides a formal route from this observation to a computational mechanism. In active inference, policy inference is precision-weighted: the same beliefs can produce decisive or tentative behavior depending on γ, the parameter controlling how sharply an agent commits to its inferred policy distribution. Prior active-inference accounts of affect and emotion link affective dynamics to model evidence, prediction error, emotional-state inference, and precision-weighted policy inference [15, 23, 14, 20]. Social-learning work similarly shows that agents revise expectations from discrepancies between expected and observed behavior of others, including in trust-game interactions [16]. However, in multi-partner settings, the evidence relevant to confidence often comes from particular partners rather than from the social environment as a whole. We therefore define partner-local affective precision from partner-response likelihood surprisal: the surprise of an observed partner response under the focal agent's current posterior over that partner's type and stance.

This mechanism complements, rather than replaces, existing accounts of social inference. Theory-of-mind models in active inference address how agents model what others know or intend through recursive belief structures and factorized generative models [21, 22]. Volatility and social-learning models address how prediction errors should change belief updating under uncertainty [18, 3, 8]. Partner-local affective precision instead estimates how strongly each current partner model should be expressed in policy. Each partner receives its own β_k estimate, updated only from that partner's behavioral evidence, which modulates policy precision when inferring policies toward that partner.

We evaluate this mechanism in a repeated multi-partner graded trust game with partner-choice conditions, abrupt betrayal, and profile-style parameter variation. The simulations test whether behavioral effects arise through policy sharpening rather than improved partner-state inference, whether partner-local precision reshapes engagement under choice and change, and whether precision gain together with prior model fitness shapes temporally structured profiles of social trust revision.

The remainder of the paper is organized as follows. Section 2 specifies the multi-partner trust game, per-partner generative models, and the partner-local affective precision mechanism. Section 3 reports simulation results. Section 4 discusses implications, limitations, and directions for future work.

2 Methods

A focal active-inference agent plays a repeated trust game with four partners whose behavioral types and stances are hidden and must be inferred from behavioral evidence. The agent is implemented using the `pymdp` library for discrete active inference [13, 6], which provides standard components for belief updating and policy selection under the active inference framework [9, 10, 19]. Partner-local affective precision is layered on top of per-partner generative models, allowing evidence and policy precision to remain relationship-specific. Full POMDP specifications, affective precision update rules, and simulation protocols are provided in Appendix A, Appendix B, and Appendix C, respectively.

2.1 Trust game

We evaluate the mechanism in a repeated multi-partner trust/investment game [4], where cumulative payoff depends on how much the focal agent invests in different partners over time. Investing more can generate larger rewards if a partner reciprocates, but larger losses if the partner defects. Success therefore depends on learning which partners are cooperative, tracking how they change over time, and calibrating investment to uncertainty.

The focal active-inference agent interacts with four partners across repeated episodes. Each partner has a latent behavioral type—cooperator, reciprocator, exploiter, or random—and a latent stance toward the focal agent—trusting, neutral, or hostile. These states are not directly observed. Instead, the focal agent must infer them from partner responses and the payoff consequences of its own actions.

On each interaction, the focal agent chooses a graded investment level $a \in \mathcal{A} = \{0, 1, 2, 3, 4, 5\}$, where 0 denotes no investment and 5 denotes maximum investment. The partner then emits a cooperate-or-defect response, $o_k^{\text{act}} \in$

$\{\text{cooperate}, \text{defect}\}$, sampled from type- and stance-conditioned cooperation probabilities (Appendix A, Table 1). With endowment $E = 10$ and return multiplier $M = 3$, focal payoff is

$$\pi(a, o_k^{\text{act}}) = \begin{cases} E - a + \frac{Ma}{2}, & o_k^{\text{act}} = \text{cooperate}, \\ E - a, & o_k^{\text{act}} = \text{defect}. \end{cases}$$

Under the default parameterization, cooperation yields $\pi(a, \text{cooperate}) = 10 + 0.5a$, whereas defection yields $\pi(a, \text{defect}) = 10 - a$. At $a = 0$, payoff is always 10, providing a risk-free baseline. Higher investment increases both potential gains from cooperation and losses from defection. The task therefore requires the agent to separate partner prediction from policy commitment: it must infer what each partner is likely to do and decide how strongly to act on that inference.

Partner responses depend jointly on latent type and stance. Cooperators and reciprocators cooperate with high probability when trusting and lower probability when hostile, whereas exploiters and random partners are less reliable overall (Appendix A, Table 1). Partner stance also evolves as a function of the focal agent's investment. Cooperative evidence strength is defined as $\rho(a) = a/(|\mathcal{A}| - 1) = a/5$, and stance transitions interpolate between low-investment/withholding and high-investment/trust-building transition dynamics (Appendix A, Table 2). Graded investment therefore affects both immediate payoff and the future social state of the interaction by continuously modulating trust-building versus trust-damaging signals.

Reported experiments use one focal active-inference agent interacting with four environment-side partners. Partners sample actions from type-by-stance cooperation probabilities and undergo action-dependent stance transitions, but do not perform variational inference or maintain affective precision estimates. This isolates the focal agent's precision mechanism before extending to reciprocal active-inference partners.

Episodes comprise 200 rounds by default and 120 rounds in abrupt-betrayal protocols, with four partners available throughout. Unless otherwise specified, partner types may drift with switching probability $p_{\text{switch}} = 0.05$.

The primary reported setting is an open graded decision regime in which the agent selects investment levels within the repeated trust game. In configurations with agent-controlled allocation, the agent chooses both a partner and an investment level, yielding a joint action space of $4 \times 6 = 24$ partner–investment combinations. In partner-choice conditions, partner selection is made explicit before investment selection, allowing approach, avoidance, and reallocation behavior to be measured directly. Graded actions are used as the primary evaluation regime because they preserve variation in policy precision, allowing partner-specific precision modulation to affect the strength of investment rather than only the direction of choice.

2.2 Per-partner generative models

Rather than maintaining a single pooled social model, the focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k. This factorization keeps evidence about each partner independent and allows policy precision to vary across relationships. Each model encodes observation likelihoods, transition dynamics, preferences, priors, and policy priors using the standard $A/B/C/D/E$ structure of discrete active inference [6]. Full matrix specifications are provided in Appendix A.

Each partner model contains three latent factors: partner type, partner stance, and a deterministic own-action bookkeeping factor that stores the executed graded investment level for the payoff likelihood. Observations are partner response and focal payoff. The partner-response likelihood is defined by type- and stance-conditioned cooperation probabilities, operationalizing predictive reliability: both a reliably cooperative and a reliably defecting partner can be predictable, even though their payoff consequences differ. Payoff enters as a second observation modality whose likelihood depends on investment level and partner response, with partner response marginalized through the cooperation likelihood. Stance transitions are graded-investment dependent: higher investment shifts stance dynamics toward trust-building transitions, while lower investment or withholding shifts them toward trust-damaging transitions.

2.3 Partner-local affective precision

Partner-local affective precision maps partner-response model fit onto policy precision. Each partner model $\mathcal{M}_k$ predicts how likely partner k is to cooperate or defect under the agent's current beliefs about that partner's type and stance. After an interaction, the observed response provides evidence about how well that partner model currently predicts the relationship. Because this update is computed separately for each partner, confidence can diverge across relationships even when the focal agent's task and preferences remain fixed.

After interacting with partner k, the agent computes

$$\epsilon_k = -\log P\big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \;\big|\; q(s_k^{\text{type}}, s_k^{\text{stance}})\big)\,. \tag{1}$$

Equation (1) is the negative log posterior predictive likelihood assigned to the observed partner response under the current type/stance posterior. Equivalently, the predictive probability marginalizes over the agent's current beliefs about partner type and stance:

$$P\big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \mid q_k\big) = \sum_{s^{\text{type}}, s^{\text{stance}}} P\big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \mid s^{\text{type}}, s^{\text{stance}}\big)\, q_k(s^{\text{type}}, s^{\text{stance}}).$$

Partner response directly tests the type-by-stance model: a reliably defecting partner can produce low partner-response likelihood surprisal despite poor payoff, while an erratic cooperative partner can produce high partner-response likelihood surprisal despite favorable outcomes. The signal therefore tracks partner-response model fit rather than partner value.

The neutral baseline is the squared surprisal of a chance-level two-outcome partner-response prediction,

$$\sigma_0^2 = (-\log 0.5)^2.$$

Because the partner response has two possible outcomes, a probability of 0.5 corresponds to a model that predicts neither cooperation nor defection better than chance. Squaring places surprisal on a nonnegative error scale before signed affective charge is computed:

$$\phi_k = \alpha\big(\sigma_0^2 - \epsilon_k^2\big)\,, \tag{2}$$

where $\alpha \geq 0$ is a gain parameter. Positive ϕ_k means partner k's response was predicted better than chance; negative ϕ_k means it was predicted worse than chance.

The signed charge ϕ_k is then used as evidence over the partner-specific inverse-precision estimate β_k. Positive charge shifts posterior mass toward lower β_k, corresponding to higher policy precision; negative charge shifts posterior mass toward higher β_k, corresponding to lower policy precision. A persistence prior smooths this update across rounds so that confidence changes gradually rather than resetting after each interaction. The full categorical update is given in Appendix B.

The agent then sets partner-specific policy precision from the posterior mean:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k], \tag{3}$$

where γ_{base} is the baseline policy precision applied in the absence of affective modulation. Because β_k represents inverse precision, higher values indicate that the partner model has been less reliable, producing a broader, less committed policy distribution. Lower values indicate stronger model fit, yielding higher γ_k and sharper policy commitment.

The β_k posterior is maintained as an auxiliary precision tracker rather than as a hidden state inside the focal agent's generative model. This keeps the mechanism lightweight and easy to isolate experimentally, but it also means the agent does not form prior expectations over its own future confidence states; this limitation is discussed in Section 4.

2.4 Cross-partner policy selection

In partner-choice regimes, the policy set is partitioned by partner. For each partner k, the set Π_k contains candidate policies involving that partner, such as different investment levels or action sequences directed toward k. The corresponding partner model $\mathcal{M}_k$ assigns each candidate policy an unnormalized policy-evidence score $u_{k,\pi}$. Partner-specific precision γ_k is then used to regulate how strongly the agent commits among policies involving that partner.

To apply γ_k locally, we separate each partner's average policy evidence from the relative differences among that partner's policies. Let

$$\bar{u}_k = \frac{1}{|\Pi_k|} \sum_{\pi \in \Pi_k} u_{k,\pi}$$

be the mean policy evidence for partner k. The precision-adjusted score is then

$$\tilde{u}_{k,\pi} = \bar{u}_k + \gamma_k \left(u_{k,\pi} - \bar{u}_k\right), \tag{4}$$

where γ_k is the partner-specific policy precision from Eq. (3). This preserves the mean evidence associated with partner k, while sharpening or flattening the differences among policies directed toward that partner. Higher γ_k amplifies those within-partner differences, producing more decisive policy commitment; lower γ_k compresses them, producing more tentative commitment.

The final policy posterior is computed by applying a softmax over the transformed scores $\tilde{u}_{k,\pi}$ across all partners and candidate policies.

2.5 Simulation setup

Simulations compare four affect-runtime variants: full partner-local precision, a shared-β ablation that pools evidence across partners while keeping per-partner POMDP beliefs intact, a tracked-only ablation that updates $q(\beta_k)$ but fixes $\gamma_k = \gamma_{\text{base}}$, and a no-affect control that disables the β tracker entirely. Reported conditions span the open graded decision regime, partner-choice regime, abrupt betrayal, precision-dynamics/profile variation, and forgiveness. Central comparisons use 30 seeds per condition; profile-scale experiments use 20 (Appendix C). Full condition definitions, episode lengths, and metrics are provided in Appendix C. Simulation code, analysis scripts, and experiment configurations will be released in a public repository upon acceptance.

3 Results

The experiments are organized around four tests of the proposed mechanism. The first asks whether partner-local precision tracks partner-response predictability rather than realized value. The second tests whether precision affects behavior through policy commitment rather than improved partner-state inference. The third asks whether partner-local precision reshapes engagement when the agent chooses whom to approach. The fourth examines how the mechanism behaves under social change, and how precision gain and prior model fitness shape the temporal structure of trust revision. Each test is evaluated by varying one element of the design while holding others fixed. We therefore treat payoff as a downstream diagnostic of how confidence calibration interacts with task structure, not as the primary criterion for whether the mechanism is present.

3.1 Partner-local precision tracks predictability rather than realized value

Partner-local precision was more strongly associated with partner-response likelihood surprisal than with realized payoff. Raw correlations showed this pattern directly ($r = -0.949$ versus $r = -0.748$). Because partner-choice episodes couple payoff, surprisal, and exposure, we also computed active-encounter partial correlations controlling for encounter count. The precision–surprisal association remained strong ($r = -0.882$), while the precision–payoff association weakened substantially ($r = 0.130$). Negative signs indicate that higher partner-response

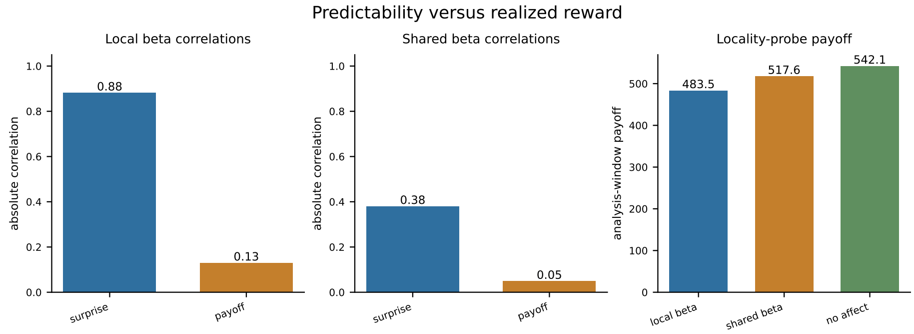

Fig. 1: Partner-local precision is more strongly associated with partner-response likelihood surprisal than with realized payoff. The correlation panels plot absolute active-encounter partial correlations; the payoff panel shows payoff over the same locality-probe analysis window. Pooling precision evidence across partners weakens the relationship between precision and partner-response likelihood surprisal, indicating that the signal depends on partner-local evidence rather than reward alone.

likelihood surprisal corresponds to lower β_k-derived policy precision; Figure 1 plots absolute magnitudes for readability.

Pooling precision evidence across partners weakened this relationship-specific signal. In the shared-β ablation, per-partner beliefs remained separate but the precision tracker was updated from pooled partner evidence. The precision–surprisal association fell to $r = -0.380$ after controlling for encounter count, and the precision–payoff association remained weak ($r = 0.049$). Thus, the update rule alone is not sufficient: precision tracks partner-response model fit most clearly when confidence evidence remains tied to the partner who generated it.

Shared-β and no-affect variants matched or exceeded partner-local precision in payoff over the locality-probe analysis window (local 483.5, shared 517.6, no-affect 542.1), despite lacking the same relationship-local precision signal. The payoff panel reports cumulative payoff over the active-encounter analysis window used for the locality probe, not full-episode payoff. Partner-local affective precision therefore should not be interpreted as reward optimization. It is better understood as model calibration: a partner-specific estimate of how confidently the agent should deploy its current model of that partner.

3.2 Tracked-only precision isolates policy commitment

Does partner-local affective precision change behavior by sharpening how existing partner beliefs are deployed, or by improving partner-state inference itself? The tracked-only ablation separates these possibilities by updating $q(\beta_k)$ normally while holding $\gamma_k = \gamma_{\text{base}}$. This leaves the precision estimate intact but prevents it from modulating policy selection.

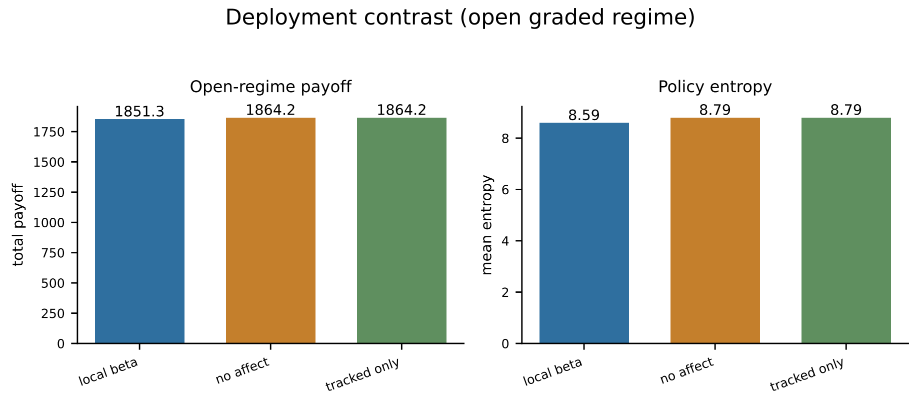

Fig. 2: Open graded regime: partner-local affective precision lowers policy entropy relative to no-affect and tracked-only controls without a corresponding payoff advantage. Tracked-only updates β_k but fixes γ_k; its readouts match the no-affect baseline, isolating the entropy effect to the $\beta_k \to \gamma_k$ policy-precision pathway.

The tracked-only ablation matched the no-affect baseline. In the open graded regime, both variants produced the same entropy (8.79 versus 8.79) and payoff (1864.2 versus 1864.2). Full precision modulation was the only variant that lowered entropy (8.59), with slightly lower cumulative payoff (1851.3). The entropy gap is modest, but its locus is clear: β_k matters behaviorally when it is allowed to modulate γ_k, producing sharper policy commitment. When that pathway is cut, the behavioral effect disappears. Figure 2 shows the three-way contrast.

This supports the interpretation of affective precision as a calibration layer rather than an inference-improvement mechanism. The tracker does not directly improve partner-state inference; it regulates how strongly the agent commits to policies based on its current partner model. Whether sharper commitment improves payoff depends on the social context rather than on precision alone.

3.3 Partner selection also sharpens under precision modulation

Partner-local precision also affected partner selection. In the partner-choice regime, the agent first selected which partner to engage and then selected an investment toward that partner. Precision modulation lowered policy entropy in this setting as well (3.99 versus 4.83), indicating that the $\beta_k \to \gamma_k$ pathway shapes partner-action policies, not only investment policies within a fixed interaction.

The allocation shift was small but consistent with sharper policy commitment: the partner-local agent moved slightly toward the cooperator (36.6% versus 34.8%) and away from the exploiter (13.8% versus 16.2%). Aggregate payoff did not meaningfully separate at this scale (393.6 versus 393.2). This reinforces the interpretation from Sections 3.1 and 3.2: partner-local precision changes how confidently the agent deploys its current partner models, but does not guarantee reward improvement. Whether sharper commitment helps or hurts depends on whether the social environment remains stable. The following abrupt-betrayal condition tests this boundary directly.

3.4 Abrupt betrayal tests confidence under change

Accumulated confidence is useful only while it remains attached to a valid partner model. The abrupt-betrayal condition tests this boundary directly: a previously cooperative partner switches stance at round 31, and the agent must adapt while carrying confidence built from the pre-switch relationship.

Precision modulation continued to affect deployment after the switch. Mean policy entropy was lower under partner-local precision than under the no-affect control (8.36 versus 8.74), with a bootstrap interval that did not cross zero (-0.62 to -0.14). Joint partner-type accuracy was also higher under precision modulation (0.372 versus 0.266; bootstrap interval 0.034 to 0.185), consistent with engagement being redirected rather than simply disrupted.

Payoff showed a weaker and more variable pattern. The payoff difference was positive but uncertain (1185.9 versus 1172.1; bootstrap interval -25.2 to 53.2). This is the expected boundary condition for a calibration mechanism: confidence can keep policies decisive after a relationship changes, but reward depends on whether the agent redirects engagement quickly enough and toward the right alternatives. The mechanism reshapes deployment reliably; its payoff consequences depend on the reallocation dynamics of the social environment. Figure 3 summarizes the post-switch readouts. The diagnostic panel reports affect-minus-control differences in joint partner-type accuracy, late post-switch encounters with P0, and wrong-type assignments; these are included as confirmation readouts rather than primary outcome measures.

The betrayal result therefore exposes a timing problem in social confidence: confidence built from reliable past evidence can remain behaviorally active after the relationship changes. We next vary precision gain and prior model fitness to examine how these confidence-revision dynamics differ across agents.

3.5 Precision dynamics as computational profiles of social trust calibration

The betrayal result shows that confidence revision has temporal structure: confidence can be too weak, too slow, or too reactive depending on how the agent updates β_k. We therefore vary precision gain α and prior model fitness to test whether trust calibration behaves like a single monotonic axis or a set of competing tradeoffs.

Sweeping α confirms that gain controls the amplitude of β_k dynamics. In the betrayal regime, mean β_k range increases from 0.147 at $\alpha = 0.05$ to 1.218 at $\alpha = 8.0$. Payoff does not follow this monotonic pattern: across open graded and betrayal regimes, higher gain reshapes confidence dynamics without producing a stable reward ranking. Faster confidence revision is therefore not automatically better confidence revision.

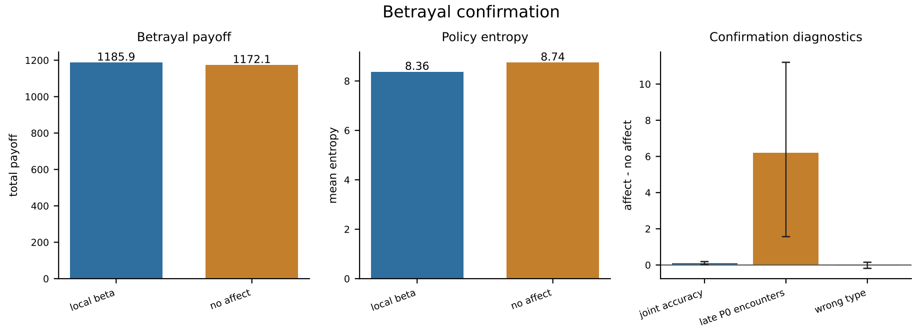

Fig. 3: Abrupt betrayal in the partner-choice regime. Partner-local affective precision lowers policy entropy relative to the no-affect control and increases joint partner-type accuracy, while the payoff difference remains uncertain. The result supports the interpretation of affective precision as a calibration mechanism whose deployment effects are reliable, but whose payoff consequences depend on post-change reallocation dynamics.

Crossing prior model fitness with gain produces distinct trust-calibration profiles. Low-gain agents show muted confidence dynamics and weaker policy commitment; high-gain agents revise more strongly but are more vulnerable to overcommitting after change. Cautious priors suppress premature confidence, but can also prevent the agent from exploiting reliably cooperative partners. In the betrayal environment, the default profile achieved the highest payoff among the tested profiles, while naive-stubborn, avoidant-rigid, and anxious-reactive profiles each expressed different failure modes. These profiles are computational hypotheses rather than fitted psychological categories.

The forgiveness paradigm separates reengagement from restored confidence. After a betrayed partner returns to cooperative behavior, some variants reengage readily without rebuilding the same confidence dynamics, while high-gain variants show larger β_k swings without faster payoff recovery. The agent can return to a partner before it trusts them again. Together, the profile analyses show that precision gain and prior model fitness jointly shape how social confidence builds, persists, and repairs. Extended tables and diagnostics are reported in Appendix D.

4 Discussion

The results establish partner-local affective precision as a calibration mechanism for social policy commitment. The mechanism defines confidence from partner-response predictability rather than realized payoff, lost its behavioral effect when the $\beta_k \to \gamma_k$ pathway was cut, and changes deployment under abrupt partner change without guaranteeing reward improvement. These findings support the central separation of the paper: social prediction and social commitment are related but distinct problems. An agent may know what a partner is likely to do while still needing to estimate how confidently that model should guide policy.

This separation explains why payoff is not the primary criterion for the mechanism. A reliably cooperative partner and a reliably defecting partner can both generate low partner-response likelihood surprisal because both are predictable, even though their payoff consequences differ. Conversely, a favorable but inconsistent partner can remain difficult to model. Partner-local affective precision therefore does not encode partner value directly. It estimates how well the current partner model is working, then regulates how strongly that model is expressed in policy. As a computational hypothesis, this predicts that social familiarity or confidence may follow predictability rather than positive value, so agents may reengage predictable-but-harmful partners or remain uncertain around favorable-but-inconsistent partners.

The tracked-only ablation gives a direct check on the mechanism: when $q(\beta_k)$ was updated but prevented from modulating γ_k, behavior matched the no-affect baseline. Thus, the behavioral effect does not come from improved partner-state inference; it appears when partner-local confidence reaches policy selection and changes how sharply the agent commits to its current partner model.

This mechanism complements existing accounts of social inference by targeting a different part of the control loop. Active-inference accounts of affect and emotion motivate links between model fit, precision, and action [15, 23, 14, 20]. Theory-of-mind models specify richer beliefs about what another agent knows or intends [21, 22], while volatility and social-learning models regulate how strongly prediction errors should revise beliefs under uncertainty [18, 3, 8]. Partner-local affective precision instead regulates how strongly the focal agent should express each current partner model in policy. A richer social agent could combine these components: infer what partners believe, estimate confidence in each partner model, detect when that confidence has become stale, and redirect exploration or commitment accordingly.

The betrayal and profile analyses show why this extra change-detection layer matters. Partner-local precision is most useful when accumulated confidence still reflects a valid relationship. After abrupt change, the same confidence can remain behaviorally active even though the evidence that produced it came from an earlier partner state. The mechanism therefore reshapes deployment reliably, but does not by itself decide when old confidence should be discounted. Future work should add volatility or change-detection mechanisms that monitor shifts in prediction-error structure and reduce stale confidence when a previously predictable partner begins behaving unexpectedly.

The profile experiments should be interpreted as computational hypotheses about trust calibration rather than fitted psychological categories. Motivated by work on social learning, psychosis, and attachment [3, 1, 5], they show how precision gain and prior model fitness jointly shape update speed, confidence amplitude, sensitivity to change, reengagement after repair, and post-betrayal commitment errors. High gain expands confidence swings without monotonic payoff improvement; low gain stabilizes behavior but slows revision; cautious priors protect against premature confidence while suppressing confidence even toward reliable partners. These profiles therefore generate predictions about social trust calibration, not claims about stable human types without behavioral fitting.

Several extensions would make the mechanism more complete. First, β_k is currently an auxiliary tracker outside the generative model; embedding it as a hidden state would allow the agent to form expectations about its own future policy precision. Second, partner-choice policies could become explicitly exploratory after detected change, allowing the agent to sample alternatives rather than only adjusting commitment toward the current partner. Third, environment-side partners could be replaced with reciprocal active-inference agents that maintain their own beliefs, policies, and precision estimates.

The most direct empirical extension is to fit the model to trust-game time series. The simulations produce measurable quantities such as exploitation sensitivity, betrayal recovery time, reengagement latency after repair, and post-betrayal commitment errors. Fitting α and prior model-fitness parameters to individual participants would test whether the proposed profiles correspond to stable differences in social trust calibration. The simulation targets that should be validated first are the paper's core contrasts: predictability over value, deployment over inference, confidence under abrupt change, and the non-monotonic effects of gain and prior model fitness.

5 Conclusion

This paper formalized social affect as a relationship-specific metacognitive signal: an estimate of how confidently an agent should act on its current model of each partner. Implemented as partner-local affective precision, this signal updates from partner-response likelihood surprisal and modulates policy precision during selection. Across simulations, it separated social prediction from social commitment: precision tracked predictability more closely than payoff, affected behavior through policy deployment rather than improved partner-state inference, and remained sensitive to the temporal structure of trust revision.

The resulting picture is not that social confidence should simply be maximized. Precision gain and prior model fitness create different calibration profiles, each with tradeoffs in confidence accumulation, betrayal adaptation, reengagement, and repair. Volatility detection, adaptive confidence discounting, and reciprocal partner models would extend the mechanism, but the core target is now clear: to test whether human trust-game behavior shows the same separations between predictability and value, inference and deployment, and reengagement and restored confidence. Partner-local affective precision offers a computational account of social metacognition in which relationships differ not only in what an agent believes about them, but in how strongly those beliefs should guide action.

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Appendix

A Full POMDP Specification

This appendix specifies the POMDP components used by the focal agent.

A.1 Hidden state factors and observations

As described in Section 2.2, the focal agent's generative model for each partner k maintains three latent factors:

$$s_k^{\text{type}} \in \{\text{cooperator, reciprocator, exploiter, random}\} \quad \text{(latent behavioral type)} \tag{5}$$

$$s_k^{\text{stance}} \in \{\text{trusting, neutral, hostile}\} \quad \text{(latent stance)} \tag{6}$$

$$s^{\text{own}} \in \mathcal{A} = \{0, 1, 2, 3, 4, 5\} \quad \text{(executed investment level)} \tag{7}$$

The own-action state is *deterministic bookkeeping*: the agent always knows its own action, and this factor makes that knowledge available to the observation model. It is not a psychological hidden variable. The joint hidden state per partner spans $4 \times 3 \times 6 = 72$ configurations.

Observations are partner action and payoff:

$$o_k^{\text{act}} \in \{\text{cooperate, defect}\} \quad \text{(partner response)} \tag{8}$$

$$o_k^{\text{pay}} \in \{5, 6, 7, 8, 9, 10, 10.5, 11, 11.5, 12, 12.5\} \quad \text{(focal agent's realized payoff)} \tag{9}$$

A.2 Partner-action likelihood

The core of the generative model is the *partner cooperation likelihood table*, which defines the probability the partner cooperates given type and stance:

Table 1: Cooperation likelihood $P(\text{cooperate} \mid s^{\text{type}}, s^{\text{stance}})$.

Partner type	Trusting	Neutral	Hostile
Cooperator	0.95	0.80	0.55
Reciprocator	0.90	0.70	0.30
Exploiter	0.70	0.35	0.10
Random	0.60	0.50	0.35

This table is the foundation of the likelihood modality A_k^{act}. Entries define

$$P\left(o_k^{\text{act}} = \text{cooperate} \mid s_k^{\text{type}}, s_k^{\text{stance}}\right), \tag{10}$$

with $P(o_k^{\text{act}} = \text{defect} \mid \cdot) = 1 - P(o_k^{\text{act}} = \text{cooperate} \mid \cdot)$. The modality is uniform over s^{own}: partner action depends only on inferred type and stance. This operationalizes what the agent considers a predictable partner—a cooperative reciprocator in a trusting stance has high predicted cooperation; an exploiter in a hostile stance has low predicted cooperation.

A.3 Payoff likelihood

Payoff enters as a second observation modality $A_k^{\text{pay}} = P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$. For investment $a = s^{\text{own}}$, endowment $E = 10$, and multiplier $M = 3$, the deterministic payoff map is

$$\pi(a, o_k^{\text{act}}) = \begin{cases} E - a + \frac{Ma}{2}, & o_k^{\text{act}} = \text{cooperate}, \\ E - a, & o_k^{\text{act}} = \text{defect}. \end{cases} \tag{11}$$

Because the partner response is not directly controlled by the focal agent, the payoff likelihood marginalizes over the partner-response likelihood:

$$\begin{aligned} &P(o_k^{\text{pay}} = v \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}}) \\ &= \sum_{o^{\text{act}} \in \{\text{cooperate}, \text{defect}\}} \mathbb{I}\big[\pi(s^{\text{own}}, o^{\text{act}}) = v\big]\, P(o_k^{\text{act}} = o^{\text{act}} \mid s_k^{\text{type}}, s_k^{\text{stance}}). \end{aligned} \tag{12}$$

This makes higher investment beneficial when cooperation is likely and costly when defection is likely, while keeping payoff preferences inside the generative model.

A.4 Transition dynamics

Type drift (B_k^{type}). Partner type is uncontrollable and drifts slowly:

$$P(s_k^{\text{type}\prime} = j \mid s_k^{\text{type}} = i) = \begin{cases} 1 - p_{\text{switch}} & j = i, \\ p_{\text{switch}}/(K_{\text{type}} - 1) & j \neq i, \end{cases} \tag{13}$$

with $K_{\text{type}} = 4$ and default $p_{\text{switch}} = 0.05$. Scheduled betrayal scenarios set $p_{\text{switch}} = 0$.

Stance dynamics (B_k^{stance}). Stance transitions depend on the focal agent's graded investment a by interpolating between a high-investment/trust-building endpoint and a low-investment/withholding endpoint:

$$\rho(a) = \frac{a}{|\mathcal{A}| - 1} = \frac{a}{5}, \qquad B^{\text{stance}}(a) = \rho(a) B^{\text{high}} + (1 - \rho(a)) B^{\text{low}}. \tag{14}$$

High investment shifts dynamics toward trust building; low investment or withholding shifts dynamics toward trust damage; recovery from hostility is slower than collapse.

Table 2: Endpoint stance transition dynamics. The focal agent's graded investment interpolates between B^{high} and B^{low} according to Eq. (14).

High investment / trust-building endpoint B^{high}			
To \ From	Trusting	Neutral	Hostile
Trusting	0.90	0.30	0.05
Neutral	0.10	0.60	0.35
Hostile	0.00	0.10	0.60
Low investment / withholding endpoint B^{low}			
To \ From	Trusting	Neutral	Hostile
Trusting	0.10	0.05	0.02
Neutral	0.50	0.35	0.18
Hostile	0.40	0.60	0.80

These dynamics instantiate the asymmetry of trust: sustained high investment gradually builds trust, while sustained withholding rapidly damages trust; recovery from hostility is slow.

Own-action bookkeeping (B^{own}). The own-action factor updates deterministically from the executed social action:

$$P(s^{\text{own}\prime} = a' \mid s^{\text{own}}, a) = \mathbb{I}[a' = a]. \tag{15}$$

This factor is not a psychological hidden variable; it makes the agent's executed action available to the payoff likelihood in standard factorized form.

A.5 Preferences, priors, and policy priors

Observation preferences (C_k). There is no direct preference over partner actions:

$$C_k^{\text{act}} = [0,\ 0] \quad \text{over } \{\text{cooperate}, \text{defect}\}. \tag{16}$$

Payoff preferences ascend over the graded payoff support generated by Eq. (11):

$$C_k^{\text{pay}} = \text{log\_softmax}\left(\frac{[5, 6, 7, 8, 9, 10, 10.5, 11, 11.5, 12, 12.5]}{\tau}\right), \tag{17}$$

where τ is a temperature parameter controlling preference sharpness. Instrumental motivation therefore enters through payoff observations and multi-step planning rather than direct action preferences.

Initial-state priors (D_k). Type, stance, and own-action priors are

$$D_k^{\text{type}} = \left[\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right], \tag{18}$$

$$D_k^{\text{stance}} = [0.2,\ 0.6,\ 0.2], \tag{19}$$

$$D^{\text{own}} = \text{Unif}(\{0, 1, 2, 3, 4, 5\}). \tag{20}$$

Policy priors (E_k). Policy priors are uniform over admissible graded investment sequences at the configured planning horizon. In partner-choice settings, partner selection is handled outside the per-partner POMDP by evaluating partner–investment policy branches with precision-adjusted policy scores before executing the chosen investment.

A.6 Partner implementation

Partner implementation follows Section 2.1. Four environment-side partners maintain latent type and stance and sample actions from the cooperation likelihood in Table 1, undergoing investment-dependent stance transitions per Table 2. Partners do not perform variational inference or maintain affective precision estimates, isolating the focal agent's precision mechanism before extending to reciprocal active-inference partners.

B Affective Precision Update and Policy Selection

This appendix gives the β-update rule, policy-precision mapping, and partner-choice policy-selection procedure.

B.1 Categorical beta support

Each partner maintains a categorical posterior $q(\beta_k)$ over discrete inverse-precision levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$, updated each round from affective charge ϕ_k. The categorical representation keeps the inverse-precision convention explicit, bounds the precision dynamics, and allows the update to be analyzed as movement over interpretable confidence states.

B.2 Persistence prior

The update proceeds in three steps. First, a persistence prior smooths the previous posterior via a tridiagonal transition matrix:

$$p(\beta_k) \leftarrow T \cdot q_{\text{prev}}(\beta_k), \tag{21}$$

where T is tridiagonal with persistence parameter 0.8 and reflecting boundaries, encoding the prior expectation that precision evolves slowly.

B.3 Charge-based pseudo-likelihood

Second, a charge-based pseudo-likelihood maps positive charge toward high policy precision (low β) and negative charge toward low policy precision (high β):

$$\log \ell(\beta_\ell) = \phi_k \cdot \frac{1}{\beta_\ell}, \tag{22}$$

where β_ℓ ranges over the five support points. This update satisfies the intended monotonicity: positive charge reinforces lower β, negative charge reinforces higher β, and the effect scales with effective precision at each support point.

B.4 Posterior update and policy precision

Third, the posterior is normalized:

$$q(\beta_k) \propto \exp(\log \ell(\beta_\ell)) \cdot p(\beta_k). \tag{23}$$

Policy precision is then set from the posterior mean as given in Eq. (3) of the main text. The quantity β_k is inverse precision, so higher posterior mean β_k lowers policy precision γ_k, while lower posterior mean β_k raises γ_k. The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent's ability to form prior expectations over its own future affective states (discussed in Section 4).

B.5 Cross-partner policy selection

In partner-choice regimes, partner-specific precision sharpens or flattens policy commitment within a relationship while preserving the mean cross-partner evidence for engaging that partner. The transformation in Eq. (4) applies γ_k to deviations from the partner-specific policy mean, rather than to raw policy scores. The agent samples or selects policies from the combined set of transformed scores across all partners and policies. This lets γ_k control commitment among policies involving partner k without changing the average evidence for approaching partner k during cross-partner comparison.

B.6 Ablation variants

The four affect-runtime variants are defined in Section 2.5. For completeness: the no-affect control disables the β tracker entirely and fixes $\gamma_k = \gamma_{\text{base}}$; the tracked-only ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without influencing policy selection; the default partner-local variant applies the full update pipeline described in B.1–B.4 and sets γ_k from each partner's posterior mean; and the shared-β ablation maintains a single pooled β state updated from all partner evidence while keeping per-partner POMDP beliefs intact, with all partner branches inheriting precision from the shared expected β.

C Simulation Protocols and Metrics

This appendix summarizes simulation conditions, replication counts, schedules, and metrics.

C.1 Simulation conditions

All reported experiments use one focal active-inference agent with four partners ($K = 4$), per-partner generative models, graded investment actions, and environment-side partner policies (Appendix A). Conditions differ by assignment mode, scheduled partner changes, and precision/profile manipulations within the graded payoff regime.

Open graded decision regime. The graded regime uses a discretized investment level while keeping the same latent type–stance structure. This is the primary deployment-contrast setting (Section 3.2): affect, no-affect, and tracked-only variants are compared on policy entropy and cumulative payoff.

Predictability-over-value locality probe. A 100-round partner-choice graded environment with partner-local, shared-β, and no-affect variants tests whether β_k-derived policy precision tracks partner-response likelihood surprisal more strongly than realized payoff (Section 3.1). The payoff panel reports the same active-encounter analysis window used for the locality-probe correlation readouts.

Partner-choice regime. The agent first selects a partner, then selects an investment level toward that partner each round. Partner selection is where engagement reorganization and social-allocation readouts are defined (Section 3.3).

Abrupt betrayal. A partner-choice graded environment with scheduled stance betrayal on a previously cooperative partner tests post-switch reallocation, entropy, payoff, and joint accuracy (Section 3.4).

Precision-dynamics/profile variation. Computational profile variants vary β_k amplitude and stability by manipulating precision gain α and prior model fitness in open graded, betrayal, and partner-choice environments (Section 3.5; Appendix D).

Forgiveness. The trust-repair protocol extends abrupt betrayal by allowing the betrayed partner to return to a cooperative/trusting state, testing reengagement and confidence recovery after restored reliability (Section 3.5).

C.2 Seed counts and episode lengths

Unless otherwise stated in the main text, episode length is 200 rounds. The abrupt-betrayal condition uses 120 rounds because post-switch readouts dominate interpretation. Profile-scale tables use 20 seeds per condition. Central behavioral comparisons use 30 seeds per condition. Exploratory diagnostic runs used three seeds per variant and were used only for implementation validation; all main-text claims are based on the confirmation-scale or profile-scale runs reported in the relevant sections.

Representative configured scales:

Table 3: Representative episode lengths and replication counts by experiment family.

Experiment family	Setting	Rounds	Seeds
Open deployment	graded, random/open choice	200	30
Predictability-over-value locality probe	graded partner choice	100	30
Partner allocation	partner choice	200	30
Abrupt betrayal	scheduled partner-choice betrayal	120	30
Precision perturbation	graded perturbations	200	30
Profile program	sweeps, factorial, forgiveness	200	20

Supplementary materials. The supplementary repository contains the simulation environments, affect-runtime variants, experiment configuration files, analysis scripts, and figure-generation code used for the reported experiments.

C.3 Betrayal schedule

The canonical abrupt-betrayal protocol (`betrayal_choice`) uses graded payoffs, agent-chosen partners, four partners with fixed initial types and stances, and no stochastic type switching ($p_{\text{switch}} = 0$). Partner P0 begins as a cooperator in a trusting stance. At round 31, P0 undergoes a scheduled stance switch to hostile while other partners retain their initial configurations. Primary readouts are post-switch cumulative payoff, policy entropy, partner reallocation, and joint partner-type accuracy over the remaining episode.

The profile-scale betrayal environments used in the alpha sweep and prior-gain factorial apply the same single-switch structure within 200-round episodes for recovery-time and quadrant metrics. The forgiveness profile extends this schedule: rounds 1–80 cooperative baseline, rounds 81–120 betrayal on P0, rounds 121–200 reversion to cooperative/trusting on P0.

C.4 Partner-choice allocation metrics

Partner-choice readouts quantify how precision reshapes *who* the agent engages, not only how it acts toward a fixed partner.

Selection share. Per-partner selection rate is the fraction of rounds on which each partner is chosen. Partner-choice reallocation is summarized qualitatively in Section 3.3.

Selection concentration. The Gini coefficient of the partner-selection distribution summarizes how concentrated engagement is across the social context (used in the profile alpha sweep and prior-gain factorial).

Post-betrayal allocation. Post-switch windows report reallocation away from the switched partner and toward alternatives, including post-betrayal P0 selection and high-investment rates in profile analyses.

Policy entropy under choice. Mean q_π entropy in partner-choice regimes captures how sharply the agent commits to partner–action policies (reported comparatively for affect versus no-affect controls).

C.5 Entropy, payoff, and accuracy readouts

Total payoff. Cumulative focal-agent payoff over the episode (or over defined post-switch windows for betrayal analyses).

Policy entropy. Mean q_π entropy (‘q_pi_entropy‘ in logged outputs) measures policy-commitment sharpness; lower entropy indicates sharper policy selection under precision modulation.

Joint partner-type accuracy. Mean joint correctness of inferred partner type (‘mean_joint_accuracy‘) summarizes epistemic state inference; it is reported separately from payoff because precision modulation can change deployment and accuracy independently of cumulative reward.

β_k **range.** Unless otherwise specified, β_k range denotes the mean within-episode range of the posterior mean $\mathbb{E}_q[\beta_k]$, averaged across relevant partners and seeds. It is an observed precision-dynamics amplitude metric, not the fixed support of the categorical β_k variable.

Betrayal timing metrics. Recovery time and related latency readouts measure rounds after a switch until selection or payoff returns toward baseline thresholds (profile alpha-sweep, prior-gain factorial, and betrayal analyses).

C.6 Replication summary

Central behavioral comparisons in Section 3 use 30 seeds per condition. Profile-scale analyses use 20 seeds per condition.

D Extended Profile and Robustness Results

This appendix reports extended parameter sweeps, profile analyses, and robustness diagnostics that support the main-text results.

D.1 Precision-dynamics parameter variation

We first vary the amplitude and stability of β_k dynamics to test how precision dynamics affect policy commitment. These variants are computational controls rather than diagnostic categories.

Table 4: Precision-dynamics parameter variation in the graded decision regime.

Variant	Mean β_k range	Total payoff	Policy entropy
affect	1.32	1851.3	8.59
low-gain	0.24	1840.5	8.9
high-gain	1.48	1859.0	8.1
cautious-prior	1.48	1877.7	8.4

Flattening the precision signal produces the clearest behavioral change. The low-gain variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines. Expanded-dynamics variants produce wider β_k movement, but payoff does not follow range monotonically. The parameter sweeps below therefore vary gain and prior structure directly.

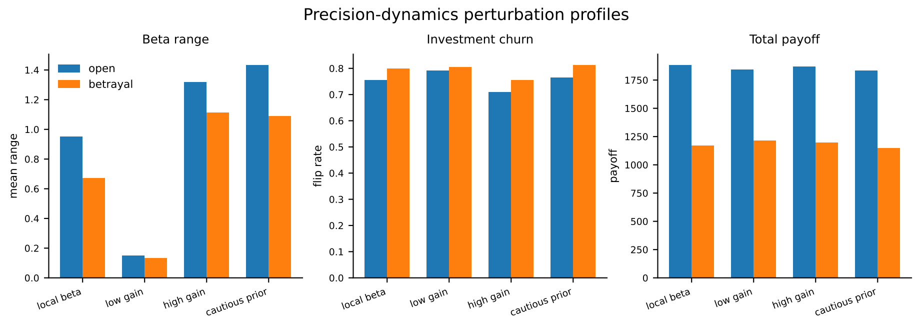

Fig. 4: Precision-dynamics parameter variation profiles in the graded decision regime. Variants compare precision-dynamics settings in the graded decision regime.

D.2 Alpha sweep

Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ tests whether precision gain controls reactive social confidence across metrics. β_k movement range increases monotonically with α in the betrayal regime:

Table 5: β_k range by precision-gain α (betrayal regime, 20 seeds).

α	Mean β_k range
0.05	0.147
0.1	0.182
0.3	0.326
0.5	0.416
1.0	0.580
2.0	0.862
4.0	1.051
8.0	1.218

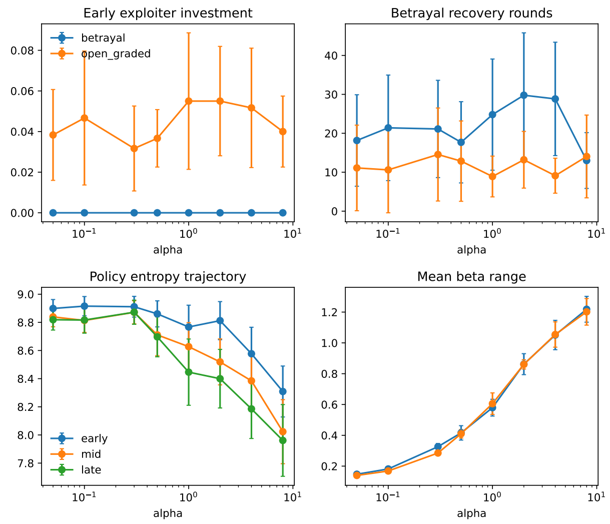

Fig. 5: Precision-gain sweep across open graded and betrayal environments (20 seeds). Higher α expands β_k range and reshapes engagement readouts without monotonic payoff ranking.

D.3 Gain-prior profile quadrants

Crossing prior model fitness with precision gain defines the profile variants used in Section 3.5. "Naive" and "cautious" refer to the initial prior over $q(\beta_k)$ on support $\{0.5, 0.67, 1.0, 1.5, 2.0\}$: naive uses $[0.40, 0.40, 0.15, 0.04, 0.01]$, while cautious uses $[0.01, 0.04, 0.15, 0.40, 0.40]$. Low and high gain refer to $\alpha = 0.1$ and $\alpha = 3.0$, respectively, with the default agent using $\alpha = 3.0$ and centered initialization at $\beta_k = 1.0$. β_k range denotes the mean within-episode range of the inverse-precision estimate. Table 6 gives key metrics.

Table 6: Behavioral metrics by profile (betrayal environment, 20 seeds). Default agent shown for reference.

Profile	Payoff	Recovery	Gini	β_k range	Trust asymmetry
Anxious-reactive	1932	26.5	0.338	1.084	0.98
Hypervigilant	1951	20.1	0.423	0.892	1.65
Naive-stubborn	1894	21.3	0.304	0.364	1.19
Avoidant-rigid	1889	14.9	0.357	0.322	1.21
Default	1991	23.0	0.423	0.893	1.68

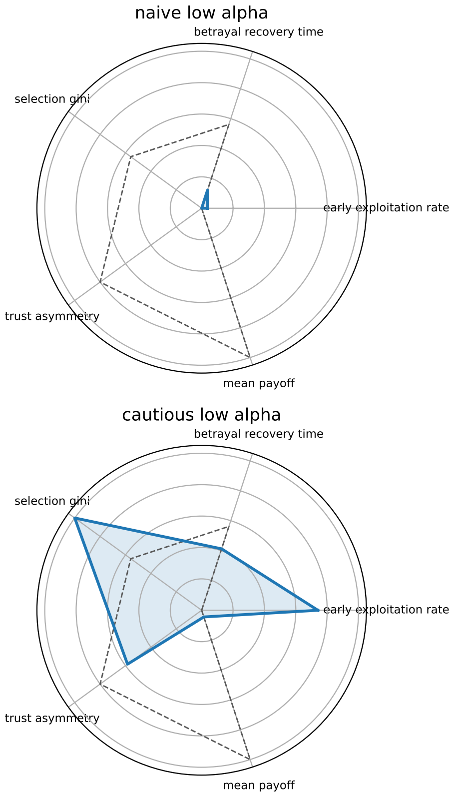

Fig. 6: Gain-prior profile quadrants from the prior-by-α factorial (betrayal environment, 20 seeds).

D.4 Forgiveness and trust repair

The forgiveness paradigm uses a 200-round partner-choice episode: cooperative baseline (rounds 1–80), betrayal on P0 (rounds 81–120), then reversion to cooperative/trusting behavior (rounds 121–200). Readouts include reengagement latency, post-return selection, payoff recovery, and β_k recovery trajectories for the reverted partner.

Table 7: Forgiveness and trust-repair metrics by precision profile (20 seeds). Reengagement is the post-repair P0 selection rate; payoff recovery compares late post-repair payoff with the late pre-betrayal baseline.

Variant	Reengagement	Payoff recovery	Latency	β_k range
cautious-high-α	0.518	0.996	5.5	0.971
cautious-low-α	0.630	1.014	1.5	0.287
default	0.475	1.019	3.0	1.012
naive-high-α	0.560	1.005	2.2	1.100
naive-low-α	0.527	1.004	4.6	0.325
no-affect	0.593	1.033	1.7	–

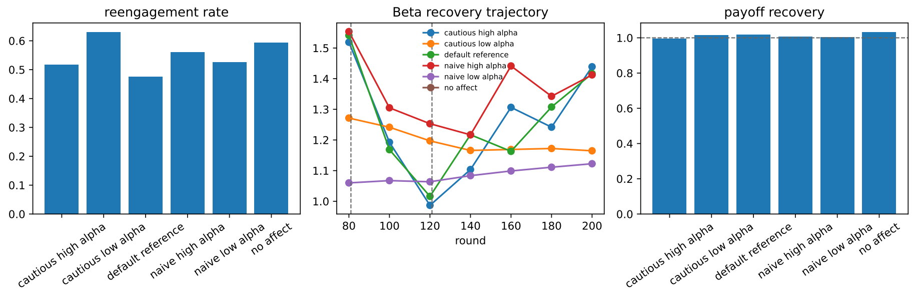

Fig. 7: Forgiveness/trust-repair diagnostic from the 20-seed profile analyses. Reengagement and payoff recovery do not collapse into a single monotonic gain effect: no-affect and cautious-low-α profiles reengage most often, while payoff recovery remains near baseline across profiles.